Keshav Sharma

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EDUCATION

Vellore Institute of Technology — **Chennai, Tamil Nadu**
Bachelor of Technology in Computer Science (Data Science Specialisation) — *Graduation: May 2027*

SKILL SUMMARY

Languages: Java, C/C++, Python, SQL
Systems Fundamentals: Operating Systems, Networking (TCP/IP), Linux/Unix, Distributed Systems, Multithreading & Concurrency, Database Management Systems (DBMS)
Reliability & Testing: Unit Testing & Test-Driven Development, Automated Testing, Crash Recovery & Fault Tolerance
Databases: MySQL, SQLite, Redis
Tools & Platforms: Git, GitHub, GitHub Actions, Docker

EXPERIENCE

VOIS – Vodafone Idea Foundation* |LINK** — ***Aug – Sep, 2025

- Applied Julius AI and LLM-based prompt engineering to analyze 2 real-world datasets (8,500+ records), extracting 15 structured business insights through natural language querying and AI-assisted data analysis.
- Iteratively refined prompts and validated AI-generated outputs for consistency, accuracy, and relevance, improving the reliability of generated insights while demonstrating effective use of AI-assisted workflows.
- Delivered a capstone project recognized with AICTE, VOIS, and Vodafone Idea Foundation certification, presenting findings to 3 mentors and translating AI-driven analysis into 4 actionable recommendations on user engagement and content trends.

PROJECTS

***RedisLite* |LINK**

- Engineered a Redis-inspired in-memory key-value database in Python with a custom RESP parser, multithreaded TCP server, thread-safe storage, O(1) LRU caching, and TTL expiration for concurrent request processing.
- Implemented Write-Ahead Logging (WAL), atomic log compaction, and automatic crash recovery to ensure durable persistence and data consistency under failures and concurrent workloads.
- Developed 46 PyTest tests covering concurrency, edge cases, and recovery scenarios; benchmarked 619K SET ops/sec in-memory and 2.8K ops/sec with WAL persistence to validate reliability and performance.

***API Rate Limiter Library* |LINK**

- Built a production-ready Python rate-limiting library implementing five throttling algorithms, Redis integration, and reusable Flask/FastAPI middleware for reliable API traffic control.
- Designed a modular architecture with pluggable storage backends and thread-safe concurrent request handling, enabling flexible deployment across application environments.
- Developed 30+ PyTest tests covering concurrency, edge cases, and algorithm correctness, and packaged the library on PyPI for public installation and reuse.

***AI-Powered Expense Receipt Scanner & Analytics Platform* |LINK**

- Built a full-stack expense management platform using FastAPI, React, SQLAlchemy, and SQLite, integrating EasyOCR, OpenCV, and NLP-based parsing to automate receipt processing and improve processing time by 40%.
- Engineered backend APIs with JWT authentication, Google OAuth, CRUD operations, analytics, budget tracking, and data export, enabling secure and reliable application workflows.
- Deployed the application across Render and Vercel, managing frontend-backend integration, authentication flows, database persistence, and production configuration for reliable end-to-end operation.

***Customer Lifetime Value Engine* |LINK**

- Developed a data processing and forecasting pipeline over 1M+ retail transactions, using BG/NBD and Gamma-Gamma models to predict 12-month customer value and churn risk for 4,000+ customers.
- Built and deployed Power BI dashboards and a Streamlit-based ROI simulator, transforming model outputs into reproducible analytics workflows for segmentation, risk monitoring, and business decision-making.

Certifications

- **Oracle Cloud Infrastructure 2025 Certified Generative AI Professional |LINK** — **2026**
  Certified in Generative AI, LLMs, Prompt Engineering, RAG, AI Agents, and OCI AI services.
- **Oracle Cloud Infrastructure 2025 Certified AI Foundations Associate |LINK** — **2026**
  Certified in AI, Machine Learning, Generative AI, LLMs, and OCI AI services.